Midterm 2, Spring 2022

This is a preview of the published version of the quiz

Started: May 9 at 2:58pm

Quiz Instructions

Question 1 — 3 pts

A type of loss-of-function allele where there is reduced function, but not complete loss

hypomorph

Question 2 — 3 pts

One gene that affects multiple traits.

pleiotropy

Question 3 — 3 pts

A set of polymorphic alleles that are inherited together

haplotype

Question 4 — 3 pts

An inversion is a type of unbalanced chromosomal rearrangement

- ○ True
- ○ False

Question 5

3 pts

An inversion homozygote will likely be sterile

- ○ True
- ○ False

Question 6

6 pts

You discover a mutant protein that is 30 amino acids longer than the wild-type protein. Provide two possible explanations for how this may have arisen by a mutation of a single base.

Edit View Insert Format Tools Table

12pt Paragraph

Many possible answers!

- base insertion that causes a frameshift
- mutation at stop codon extends protein
- mutation at splice site causes change in splicing that creates longer protein

p 0 words

Question 7

3 pts

After performing a GWAS study to find SNPs associated with diabetes, you find that most of the SNPs are in non-coding regions. To identify regions of the genome that are potential promoters, which assay could you perform? There may be multiple correct answers.

- ☒ ATAC-Seq
- ☐ ChIP-Seq on a transcription factor that binds to enhancers
- ☐ Bisulfite sequencing

Question 8

3 pts

You find that one of the SNPs is located in the promoter of a gene that is normally turned on at a low level in pancreatic islet β cells (the cell type that produces insulin). You decide to perform further experiments in islet β cells and find that one of the GWAS SNP variants, G, causes higher expression of the gene compared to the more common T allele.

Cells that are homozygous TT have low total expression of the gene, heterozygote TG have medium total expression, and homozygous GG have high expression. What type of allele is the G allele?

- Incomplete dominant loss-of-function
- Recessive loss-of-function
- Complete dominant loss-of-function
- Incomplete dominant gain-of-function
- Dominant negative gain-of-function
- Complete dominant gain-of-function

Question 9

3 pts

A disease is caused by homozygosity for the b allele (B is the corresponding wild-type allele). However, the penetrance of the disease is 25%. One individual is known to be a heterozygote and has a child with another individual that has the disease. What is the probability that the child has the disease?

- 1/16
- 1/8
- 1/4
- 1/2
- 0
- 1

Bb x bb

$\frac{1}{2}$ bb x $\frac{1}{4}$ penetrance = $\frac{1}{8}$

Question 10

3 pts

2:1 ratio suggests a lethal allele

Two blue birds were crossed. The offspring showed a 2:1 ratio of blue birds and light blue birds. What is the most likely type of inheritance that this suggests? There may be multiple correct answers.

- [ ] This is a trait controlled by two genes where the dominant allele of either of two genes is necessary to produce the blue color.
- [ ] This is a trait controlled by one gene where the dominant allele of either of two genes is necessary to produce the blue color.

- [ ] For every gene involved in the trait there is one complete dominant allele to another.
- [ ] For every gene involved in the trait there is one incomplete dominant allele to another.

Question 11

3 pts

A child has blood group O. One of his parents has blood group A and the other parent has blood group B. If these parents then have non-identical twins, what is the probability that both twins will have blood group A?

Ai x Bi

ii

	A	i
B	AB	Bi
i	Ai	ii

$\frac{1}{4}$ A

$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

twin 1 twin 2

- ( ) 1/16
- ( ) 1/8
- ( ) 1/4
- ( ) 1/2

Question 12

3 pts

Two wild-type fragments of human genomic DNA from the long arms of two nonhomologous acrocentric chromosomes are shown below. The red lines numbered 1 and 2 indicate the positions of breakpoints that could be caused by ionizing radiation. These breaks could produce chromosomal rearrangements when the broken pieces are stitched back together by DNA repair systems.

GENE A – EXON 1

```
5′-GCGATACGATGCCCGGGGTACGATCAAACGTCGATGACTAGGAATTACGCAC-3′
3′-CGCTATGCTATGGGCCCCATGCTACTTTGCAGCTACTGATCCTTAATGCGTG-5′
```

1

GENE B – EXON 3

```
5′-ATGGAGGTAGGGACCGACG-3′
3′-TACCTCCATCCCTGGCTGC-5′
```

2

What type of chromosomal rearrangement would form if the breaks were stiched back together?

- ( ) Deletion
- ( ) Duplication
- ( ) Inversion
- ( ) Translocation

Question 13

3 pts

Breakpoint 1 occurs in Exon 1 of Gene A and Breakpoint 2 occurs in Exon 3 of Gene B. In one of the chromosome products from the rearrangement, the sequence before Breakpoint 1 in Exon 1 of Gene A is stitched with the sequence of Exon 3 of Gene B after Breakpoint 2. Is there a possible protein that could be produced from this chromosome product? If so, what is the amino acid sequence?

Input your amino acid sequence using the three-letter abbreviation for each amino acid. For example "Arg Gln His". If no protein product can be made, input "No protein product".

Met Pro Gly Val Arg Ser

Stop codon is not part of the amino acid sequence, but it is fine if this was included

Question 14

3 pts

Based on your answer above, what would be the most likely resulting affect on the Gene A allele?

X Loss-of-function b/c it would be truncated

Gain-of-function

Question 15

55 pts

Extended calculations are given in class. A PDF version of the questions are here:

BME105_Spring22_Midterm2_EC_v2.pdf (https://canvas.ucsc.edu/courses/52787/files/5608893/download?download_frd=1)

You can turn in a physical copy or upload all your answers as one file here. If submitting an electronic version, you still need to turn in the top page of the physical copy version, checking the box to indicate that you've submitted electronically.

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